CHRISTOPHER B. WOMACK

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EDUCATION

2024–2026 **Ph.D.**, Aeronautics and Astronautics
Massachusetts Institute of Technology, Cambridge, MA, USA
Thesis title: *On the theory and optimal design of emulators for climate impact assessment*
Advisors: Noelle E. Selin and Glenn R. Flierl

2022–2024 **S.M.**, Technology and Policy
Massachusetts Institute of Technology, Cambridge, MA, USA

2021–2024 **S.M.**, Aeronautics and Astronautics
Massachusetts Institute of Technology, Cambridge, MA, USA

2017–2021 **B.S.**, Aerospace Engineering
Massachusetts Institute of Technology, Cambridge, MA, USA

RESEARCH EXPERIENCE

2026– **Postdoctoral Researcher**, Land-Climate Dynamics Group
ETH Zürich Institute for Atmospheric and Climate Science, Zürich, Switzerland

2023–2026 **Graduate Research Assistant**, Selin Group
MIT Center for Sustainability Science and Strategy, Cambridge, MA, USA

2023 **Computational Physics Research Intern**, Theoretical Physics Group
GSI Helmholtz Centre for Heavy Ion Research, Darmstadt, Germany

2021–2023 **Graduate Research Assistant**, Darmofal Group
MIT Aerospace Computational Design Laboratory, Cambridge, MA, USA

2019–2020 **Undergraduate Research Assistant**, Darmofal Group
MIT Aerospace Computational Design Laboratory, Cambridge, MA, USA

TEACHING EXPERIENCE

2026– **Group Project Supervisor**, Land-Climate Dynamics
ETH Zürich Institute for Atmospheric and Climate Science, Zürich, Switzerland

2026 **Co-Instructor**, Fundamentals of Climate Science and Policy
MIT Center for Sustainability Science and Strategy, Cambridge, MA, USA

2025–2026 **Teaching Certificate**, Kaufman Teaching Certificate Program
MIT Teaching + Learning Lab, Cambridge, MA, USA

2025–2026 **Lead Instructor**, Quantitative Methods
MIT Technology and Policy Program, Cambridge, MA, USA

2021–2022 **Teaching Assistant**, Introduction to Computational Science and Engineering
MIT Department of Aeronautics and Astronautics, Cambridge, MA, USA

2020 **Lead Instructor**, Electricity and Magnetism
Escola Andorrana de Batxillerat, Andorra la Vella, Andorra

2019 **Lead Instructor**, Aerospace Physics
Nazarbayev Intellectual School, Karaganda, Kazakhstan

SUPERVISORY EXPERIENCE

Dates shown correspond to the year(s) of advising. Surnames of minors are omitted for privacy.

2025–2026 **Rinoa Oliver**, B.S., Undergraduate Research Assistant
Massachusetts Institute of Technology, Cambridge, MA, USA

2025 **Kofi Ocran**, B.S., Undergraduate Research Assistant
Massachusetts Institute of Technology, Cambridge, MA, USA

2025 **Lou Lahn**, B.S., Undergraduate Research Assistant
Massachusetts Institute of Technology, Cambridge, MA, USA

2024 **Lauren Higgins**, B.S., Undergraduate Research Assistant
Massachusetts Institute of Technology, Cambridge, MA, USA

2023 **Dev**, High School Research Assistant
Polygence Research Program

2023 **Marshall**, High School Research Assistant
Polygence Research Program

SERVICE, OUTREACH, AND LEADERSHIP ACTIVITIES

2026– **Expert Reviewer**, *Schmidt Sciences*
- Evaluate and score early-stage research proposals to advise on multi-million-dollar funding allocations

2024–2026 **Local Science Partner**, *American Geophysical Union*
- Advocated for science-informed policy by building relationships with federal legislators to advance science funding and legislative priorities

2026 **Co-Convener**, *CMIP Community Workshop*
- Co-convened the first-ever session on advanced diagnostics, emulation, and machine learning for CMIP7

2022–2026 **Team Whip**, *MIT Ballroom Dance Team*
- Managed registration, space coordination, and $15,000+ per semester in transactions for 80+ team members

2022–2026 **Graduate Resident Advisor**, *Burton-Conner House*
- Provided live-in mentorship to over 50 undergraduate students, assisting in handling emergency situations, including mental health crises, and facilitating access to campus resources

2025 **Presenter**, *New York Climate Week Innovation Showcase*
- Represented MIT to industry stakeholders by presenting on the emulation of spatially explicit climate outcomes; proposal selected via competitive internal application

2025 **Workshop Facilitator**, *World Organization for Sustainability Leadership*
- Led lectures and mentored high school students in the UN's Junior Ambassadors for Sustainable Development program, guiding the development of projects on Nationally Determined Contributions

2024–2025 **Representative**, *MIT Graduate Student Council*
- Represented graduate students by attending council meetings, engaging with constituents, and voting on funding allocations and resolutions to address student needs

2024 **Presenter**, *Cambridge Science Festival*
- Developed and presented accessible climate visualizations for public outreach, educating community members on climate science

2022 **Social Chair**, *Graduate Association of Aeronautics and Astronautics*
- Managed a $25,000 budget and coordinated with faculty to plan and execute events for over 250 graduate students

2022 **Congressional Visit Days**, *MIT Science Policy Initiative*
- Advocated for increased federal funding in basic research by engaging in over 30 meetings with congressional offices; final report secured program funding for the subsequent year

PUBLICATIONS, TALKS, AND SOFTWARE

Peer-Reviewed Journal Articles

[1] **Womack, C. B.**, Bouabid, S., Sokolov, A., Salunke, P., Flierl, G., Eastham, S. D., & Selin, N. E. (in press). Optimal scenario design for climate emulation. *Proceedings of the National Academy of Sciences*. arXiv:2606.19302.

[2] Bouabid, S., **Womack, C. B.**, Selin, N. E., & Ferrari, R. (2026). Evidence that end-of-century emulation errors under overshoot remain largely within internal variability in an Earth system model. *Environmental Research Letters, 21*(16), 164019.

[3] **Womack, C. B.**, Flierl, G., Bouabid, S., Souza, A. N., Giani, P., Eastham, S. D., & Selin, N. E. (2026). A theoretical framework to understand sources of error in Earth System Model emulation. *Earth System Dynamics*, 17(1), 107–139.

[4] **Womack, C. B.**, Giani, P., Eastham, S. D., & Selin, N. E. (2025). Rapid emulation of spatially resolved temperature response to effective radiative forcing. *Journal of Advances in Modeling Earth Systems*, 17(1), e2024MS004523.

Manuscripts Under Review, In Prep, and Preprints

[1] **Womack, C. B.**, Arrabali, D. M., Quilcaille, Y., Gudmundsson, L., Seneviratne, S. I. (in prep). Targeted climate memory improves regional overshoot emulation where hysteresis emerges above internal variability.

[2] Schöngart, S., Gudmundsson, L., **Womack, C. B.**, Schleussner, C.-F., & Seneviratne, S. I. (in revision, *Science Advances*). A review of spatially explicit climate emulators for enhancing modelling agility. `https://www.researchsquare.com/article/rs-11205768/v1`.

[3] Schöngart, S., Sandstad, M., **Womack, C. B.**, . . . & Seneviratne, S. I. (submitted, *Earth System Dynamics*). FASTMIP-CMIP7: A Coordinated Multi-Emulator Ensemble for Rapid, Spatially Resolved Climate Projections under CMIP7 Scenarios.

[4] **Womack, C. B.**, Bouabid, S., Gao, X., Morris, J., Schlosser, C. A., Flierl, G., & Selin, N. E. (preprint). Assessing spatially explicit sensitivities to scenario uncertainty through climate emulation. *ESS Open Archive*. `https://doi.org/10.22541/essoar.15008111/v1`.

Invited Talks

[1] **Womack, C. B.** (2026). Advancing climate emulator generalization through optimized training data and the FASTMIP emulator intercomparison. *Allen Institute for AI.*

[2] **Womack, C. B.** (2026). On the theory and optimal design of emulators for climate impact assessment. *ETH Zürich Extraordinary Seminar.*

[3] Roy, E., **Womack, C. B.**, Finkel, J., & Flierl, G. (2025). Charting future Pathways: Simple Climate Models. *WOSL, UN Junior Ambassador for Sustainable Development Seminar.*

[4] **Womack, C. B.**, Giani, P., Eastham, S. D., Flierl, G., & Selin, N. E. (2024). Applications of Response Functions for Rapid Climate Emulation. *Met Office Simple Climate Model Seminar.*

Conference Presentations and Papers

[1] Bouabid, S., **Womack, C. B.**, Flierl, G., Selin, N. E., Ferrari, R., Souza, A. N., . . . & Lütjens, B. (2026). Advances in generative climate emulation to support impact-assessment. *EGU General Assembly 2026* (EGU26-10825).

[2] Schöngart, S., Gudmundsson, L., **Womack, C. B.**, Schleussner, C.-F., & Seneviratne, S. I. (2026). A review of spatially explicit climate emulators for enhancing modelling agility. *EGU General Assembly 2026* (EGU26-6394).

[3] **Womack, C. B.**, Bouabid, S., Sokolov, A., Salunke, P., Flierl, G., Eastham, S. D., & Selin, N. E. (2026). Optimal scenario design for climate emulation: How to train your emulator. *CMIP Community Workshop 2026.*

[4] **Womack, C. B.**, Flierl, G., Bouabid, S., Souza, A. N., Giani, P., Eastham, S. D., & Selin, N. E. (2025, December). Diagnosing and Comparing Sources of Error in Emulators of Earth System Models. In *AGU Fall Meeting Abstracts* (Vol. 2025, No. 855, pp. GC43F-0855).

[5] Selin, N. E., Ferrari, R., Bouabid, S., Chen, Y. H. H., Fiore, A. M., Flierl, G., . . . & **Womack, C. B.** (2025, December). Advancing Emulation for Understanding Nature-Society Interactions. In *AGU Fall Meeting Abstracts* (Vol. 2025, pp. GC51A-01).

[6] **Womack, C. B.**, Flierl, G., Wang, R., Souza, A. N., Ferrari, R., & Selin, N. E. (2024, December). Accounting for Memory and Explainability in Emulating Dynamical Systems. In *AGU Fall Meeting Abstracts* (Vol. 2024, pp. GC13V-07).

[7] **Womack, C. B.**, Selin, N. E., & Eastham, S. D. (2024). Rapid Emulation of Spatially Resolved Temperature Response Functions to Effective Radiative Forcing. *EGU General Assembly 2024* (EGU24-18412).

[8] Schroeder, M., **Womack, C. B.**, & Gagnon, A. (2020). Maneuver planning for demonstration of a low-thrust electric propulsion system. *Small Satellite Conference, Technical Session VII: Student Competition.*

Software

[1] **Womack, C. B.** (2025). Source code for "A theoretical framework to understand sources of error in Earth System Model emulation." *Zenodo*.

[2] **Womack, C. B.** (2024). Source code for "Rapid emulation of spatially resolved temperature response to effective radiative forcing." *Zenodo*.

Theses

[1] **Womack, C. B.** (2026). *On the theory and optimal design of emulators for climate impact assessment.* (Doctoral thesis, Massachusetts Institute of Technology).

[2] **Womack, C. B.** (2024). *Development and Stakeholder-informed Evaluation of Global Climate Temperature Response Functions.* (Master's thesis, Massachusetts Institute of Technology).

[3] **Womack, C. B.** (2024). *On the Potential Impact of Curved Meshing for Higher-order Adaptive Mesh Simulations.* (Master's thesis, Massachusetts Institute of Technology).

COMPETITIVE FUNDING

2026 **Advanced Study Program Postdoctoral Fellowship**, *declined*
National Center for Atmospheric Research

2026 **CMIP Community Workshop Travel Support Award**
CMIP International Project Office

2025 **Conference Grant**
MIT Office of Graduate Education

2025 **MIT New York Climate Week Travel Grant**
MIT Climate Project

2025 **Dynamical Core Intercomparison Project Travel Grant**
National Center for Atmospheric Research and University of Michigan

2025 **Congressional Visit Days Travel Grant**
American Geophysical Union

2023 **RISE Professional Fellowship**
Deutscher Akademischer Austauschdienst

AWARDS AND HONORS

2025 **Finalist**
MIT Research Slam

2022 **Outstanding Leadership Award**
MIT Department of Aeronautics and Astronautics

2021 **Henry Webb Salisbury Award for Student Leadership**
MIT Department of Aeronautics and Astronautics

2020 **Student Paper Competition Honorable Mention**
Small Satellite Conference 2020

ADDITIONAL

Programming Languages: Python, Julia, C++
Tools: MATLAB, Version Control, JAX
Languages: English (native), German (A2)
Interests: Backpacking, Competitive Video Games, Film Discussion and Review